Why axe Monitor and SiteImprove disagree

Reconciling the two automated accessibility scores across the NYS site portfolio.

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This note explains why the same website earns a much higher accessibility score in SiteImprove than in axe Monitor — across the sites we scan with both tools, about **37 points higher on average** — and how the dashboard handles that gap when it reports and compares sites.

Bottom line

The two tools are **not two readings of the same measurement.** They use different formulas that count different things:

- **axe Monitor** scores **whole pages** by their single worst problem, then averages.
- **SiteImprove** scores **individual checks** — what share of everything it tested passed.

A large, repeatable gap between them is **expected**, not a sign that either tool is broken.

Do not "add 37 points" to convert one score into the other. The gap is not a fixed constant. In our own data it ranges from roughly **–3 to +84 points**, and it *widens on our worst sites and shrinks on our cleanest* — the opposite of what a fixed offset assumes. A "+37" crosswalk would quietly forgive exactly the sites with the most serious blockers.

For the compliance question this dashboard exists to answer — "*what share of our pages are free of serious blockers?*" — **axe Monitor is the defensible lens**, and it is what drives the status color. SiteImprove is reported alongside it, not blended into it.

1. They don't measure the same thing

It is tempting to treat these as two thermometers that disagree, as if one is running hot. That framing is the mistake. They are two different instruments measuring two different quantities.

- axe Monitor asks a strict, page-by-page question: **"What share of our pages are free of high-severity problems?"**
- SiteImprove asks a volume question: **"Of all the individual checks we ran, what share passed?"**

Both are reasonable. They answer to different bosses, so their numbers will not line up — and the one most people already have access to (SiteImprove) is the optimistic one.

2. How each tool actually scores

axe Monitor — "worst problem wins"

axe Monitor finds the single most severe issue on each page and sorts the whole page into one bucket based on that one issue, then averages across pages:

Page's worst issue	Page counts as
Minor only, or none (**Good**)	100%
Moderate (**Fair**)	80%
Serious	40%
Any Critical issue	**0%** — a hard zero, no partial credit

Two consequences matter enormously:

1. **The number of problems on a page barely moves the score.** A page with one serious issue and
   a page with four hundred serious issues both score 40%.
2. **A single critical issue erases the page entirely.**

For our templated state sites this is decisive. The shared header, footer, and navigation appear on every page. If any one shared component carries a serious issue — a low-contrast link, an unlabeled menu button — then **every page inherits it and caps at 40%.** That is why our axe scores cluster in a low, narrow band: **40% of the sites we scan with both tools score between 40 and 55.** The formula is punishing by design.

SiteImprove — "credit for everything that passes"

SiteImprove works at the level of individual checks. A single page runs through thousands of automated checks, and the overwhelming majority pass. Three features push the number high:

- **The base is enormous.** Every element that passes is counted, so a handful of failures is diluted across thousands of passes.
- **No page-level collapse.** One bad element costs one check; the rest of the page keeps its credit.
- **Unconfirmed items wait offstage.** Findings that need human confirmation are held as "potential issues" and don't drag the automated score down until reviewed.

Side by side

	axe Monitor	SiteImprove
What it counts	Whole pages	Individual checks (occurrences)
How a page is judged	By its single worst issue	Every passing check earns credit
Effect of one serious issue	Caps the whole page at 40%	Loses credit for that one check
Effect of a critical issue	Forces the page to 0%	One more failed check among thousands
Denominator	Number of pages (small)	Number of checks run (very large)
Typical result	Clusters low	Runs high

3. What our data shows (snapshot 2026-07-15)

Of **292 sites**, **94** are scanned by both tools. On those 94:

	axe Monitor	SiteImprove (Accessibility)
Average score	**53.7**	**90.3**

	axe Monitor	SiteImprove (Accessibility)
Higher score	1 of 94 sites	**93 of 94 sites**

- **Average gap: +36.6 points** in SiteImprove's favor.
- SiteImprove places **89 of the 94** shared sites in the green band (80–100) where axe Monitor does **not** — and **15 of those are axe-red** (0–40).

Representative sites:

Site	axe Monitor	SiteImprove
agriculture.ny.gov	53	90
arts.ny.gov	39	88
apa.ny.gov	43	73

4. Why the gap is steady — and why "+37" is a trap

A gap that hovers near 37 points across many sites is the fingerprint of a **structural formula difference**, not random noise. Most of our sites share the same header and footer, so they land in the same axe Monitor bucket (Serious) across the board — pinning those scores in a tight low band while SiteImprove's share-of-checks floats high.

The temptation is to "normalize" by adding a fixed number to the axe scores. **Do not.** A site with a genuine critical blocker craters in axe Monitor (a hard zero for that page) while barely moving in SiteImprove (one more failed check among thousands). So the gap **widens precisely on our worst sites and narrows on our cleanest** — and our data bears this out:

	Average gap (SiteImprove – axe)
Our 10 **worst** axe sites	**+65.1 points**
Our 10 **best** axe sites	**+13.2 points**

A "+37" crosswalk would apply the smallest correction where it is needed most, quietly forgiving the sites with the most serious blockers.

5. Before comparing the two columns

Two checks keep this an apples-to-apples comparison. **The dashboard already handles both:**

- **Compare the right SiteImprove number.** SiteImprove's headline DCI blends three equally-weighted modules — Accessibility, Quality Assurance, and SEO — and QA/SEO run high.
  Comparing that headline against axe Monitor's accessibility percentage would inflate the gap further. The pipeline reads SiteImprove's **Accessibility sub-score** (`a11y.total`) specifically, never the blended DCI headline — so the gap above is a like-for-like, accessibility-only comparison.
- **Confirm the crawl scope.** Because axe Monitor scores *pages*, its number swings with which pages and how many were scanned; a shallow crawl versus a deep one can manufacture a gap on its
  own. The dashboard now surfaces both tools' page counts side by side — **"Pages tested"** (axe Monitor) and **"Pages indexed"** (SiteImprove) — so a scope mismatch is visible per site rather than hidden. Notably, axe's lower score is **not** an artifact of a thin sample: on many sites it crawls *more* pages than SiteImprove indexes (e.g. apa.ny.gov — 953 tested vs. 153 indexed) and still scores lower.

6. What this means for the dashboard

- **We report both numbers, labeled.** SiteImprove answers "*how many defects do we have, and are*
  we reducing them?" — the better day-to-day remediation metric. axe Monitor answers "*what share*
  of pages are free of serious blockers?" — the stricter compliance lens. Used together they are complementary; collapsed into one number they mislead.
- **Status color follows axe Monitor.** Because the dashboard's job is a compliance early-warning, the single status chip uses the authoritative axe Monitor score when present (SiteImprove only where axe has none). A `blocked` flag forces Red regardless of score.
- **Automated ≠ compliant.** Automated tools catch roughly **30%** of accessibility barriers. Even the stricter axe Monitor number is a *floor* — the real compliance picture is at best what axe shows and likely worse. Manual auditing remains essential; that is what the Auditor column is for.
- **Known coverage gap. 27 sites** are currently scored by SiteImprove only. Their status rests on the softer number until they are added to axe Monitor scanning (tracked as a separate operational follow-up).

7. DCT Council talking points

- **Two tools, two formulas — not two readings of one thing.** SiteImprove counts individual checks
  (a huge denominator, so scores run high); axe Monitor scores whole pages by their single worst issue (one serious problem in a shared header caps the page at 40%).
- **The rosy number is the one most people see.** Across our 94 dual-scanned sites SiteImprove averages ~37 points higher and rates 89 of 94 "green" where axe Monitor does not.
- **You cannot adjust the gap away.** It is ~65 points on our worst sites and ~13 on our cleanest; a fixed offset would hide the worst offenders.
- **For the compliance question, axe Monitor is the defensible metric** — and it drives the dashboard's status.
- **Even axe is a floor.** Automation catches ~30% of barriers; the true picture requires manual review.

Appendix: the formulas

Included for the record. Not required to follow the argument above.

axe Monitor

Each page is bucketed by its single worst issue, then:

$$\text{Score} = (0.4 \cdot p_2 + 0.8 \cdot p_1 + 1.0 \cdot p_0) / TP$$

where p_2 = number of Serious pages, p_1 = number of Fair pages (worst issue Moderate), p_0 = number of Good pages (Minor only / none), and TP = total pages. Any page containing a Critical issue contributes 0 to the numerator. The overall score is the average of the per-page scores. Verified against Deque's documentation (see Sources).

The severity tiers (Critical / Serious / Moderate / Minor) come from axe-core and reflect Deque's assessment of user impact. They are **not** defined by WCAG; WCAG does not rank its success criteria by severity.

SiteImprove

The Site Target Score — the cleanest documented form of the ratio — is:

```
Score = 1 − (failed occurrences / total occurrences detected)
```

evaluated over the success criteria in the chosen WCAG target. The **DCI Accessibility sub-score** that the dashboard reads (`a11y.total`) is a weighted algorithm over how many WCAG success criteria
the site satisfies across A / AA / AAA plus WAI-ARIA and best practices, yielding a 1–100 value; it lands lower than the raw Site Target Score, typically in the 75–95 range. The blended DCI headline (which the dashboard does **not** use) is:

```
DCI = (Accessibility + Quality Assurance + SEO) / 3
```

How to refresh the numbers in this note

The snapshot figures (Sections 3–4) are point-in-time, computed from `public/dashboard-data.json` . Regenerate that file with `npm run generate` (or `npm run generate:offline` from cache), then recompute the overlap statistics against the `sites` array. Update the snapshot date at the top when you do.

Sources

- axe Monitor — Scan Overview (score formula): https://docs.deque.com/monitor/8.3/en/scan-overview/
- axe-core — Issue Impact definitions: https://github.com/dequelabs/axe-core/blob/develop/doc/issue_impact.md
- SiteImprove — Digital Certainty Index (DCI): https://help.siteimprove.com/support/solutions/articles/80000448555
- SiteImprove — Accessibility Site Target Score: https://help.siteimprove.com/support/solutions/articles/80001152008